



10

TUTORIAL TEN

BUDGETING, PLANNING AND CONTROL

Mini Case: Armadillo Surfing Design (ASD)

Adapted from "An Introduction to Accounting: Accountability in Organisations and Society", 1st edition, Cengage

THE CASE:

In addition to surfboards and wetsuits, ASD now stocks a full range of men's and women's apparel including hoodies, tracksuit pants, and swimwear.

ASD's market research and data analysis indicate that approximately 85% of sales are from their men's lines and just 15% of sales are from their female lines. They are keen to broaden their appeal to the female market and have decided to develop a beach towel line which they will manufacture themselves and market primarily to women.



Maddie will be managing the beach towel project and has asked you to meet with her to discuss some of the key details including the expected sales, the manufacturing costs, and the bonus structure for manufacturing staff, including the production manager. At the meeting, you make the following notes:

Armadillo Surf Designs

Meeting Notes: 6 February 20X4

Subject: Beach Towel Range

- Sales price to be \$85
- Annual sales – 12,000 units (15% of sales in July to Sep, 35% of sales in Oct to Dec, 35% of sales in January to March, 15% of sales in April to June)
- Important not to run out of stock. Aim for 400 beach towels in stock at the end of each quarter. Manufacturing has not yet commenced and so starting inventory is nil.
- Materials will cost \$22 per towel. It is important not to run out of materials. The fringing is imported from Europe and takes up to 35 days to arrive so we must keep back-up stock of this on hand to ensure we do not run out or manufacturing will be interrupted. There is currently \$8,000 of materials (including fringing) on hand and the aim is to have this amount of material remaining at the end of each quarter.
- The towel fabric comes pre-printed. On average, it takes each worker 6 minutes to cut the towel to size, 6 minutes to attach the fringing and 3 minutes to attach the labels and tags and package the towel. Labour costs (including on-costs such as superannuation and payroll tax) are \$32 per hour. Manufacturing staff to receive bonus if they exceed their quarterly production targets. No bonuses are expected to be paid for at least 18 months.
- The machinery to be used in the factory will be leased at a cost of \$8,500 per month. The lease agreement states that ASD is responsible for servicing and maintenance of the equipment which is expected to cost \$0.25 per unit produced. The electricity to run the machinery is expected to cost \$0.75 per unit produced.
- A production manager has been employed to supervise the production process and will be paid a salary (including on-costs) of \$85,000 per year and will receive a bonus if the labour and/or material cost per beach towel is reduced. It is considered that such a bonus is unlikely to be paid in the next financial year.

ACTIVITIES

1. From the information obtained during the meeting, prepare the following operating budgets (with a quarterly breakdown) for the new beach towel line for the year ending 30 June 20X5. a)

Sales Budget

b) Production Budget

c) Material Budget

d) Labour Budget

e) Overhead Budget



Solution

Operating Budgets for Armadillo Surf Designs

Armadillo Surf Designs Sales Budget for the year ending 30 June 20X5					
	Q1	Q2	Q3	Q4	Total
Target sales (units)					
Selling price					
Total sales revenue					
Armadillo Surf Designs Production Budget for the year ending 30 June 20X5					
	Q1	Q2	Q3	Q4	Total
Expected sales in units					
Add: Desired closing inventory					
Deduct: Opening inventory					
Required production in units					
Armadillo Surf Designs Material Budget for the year ending 30 June 20X5					
	Q1	Q2	Q3	Q4	Total
Units to be produced					
Material cost per unit					
Cost of material required for production					
Target ending inventory of materials					
Total materials required					
Less opening inventory of materials					
Total cost of materials purchases					

Solution

Armadillo Surf Designs Labour Budget for the year ending 30 June 20X5

	Q1	Q2	Q3	Q4	Total
Units to be produced					
Labour time per unit (in hours)					
Total labour hours required					
Labour cost per hour					
Total labour cost					

Armadillo Surf Designs Overhead Budget for the year ending 30 June 20X5

	Q1	Q2	Q3	Q4	Total
Units to be produced					
<u>Variable Costs</u>					
Maintenance					
Electricity					
<u>Fixed Costs</u>					
Machinery lease					
Indirect labour					
Total production overhead					

Multiple Choice Questions

1. Which statement about a budget is correct?

- A) The master budget expresses management's operating and financial plans for a specified period (usually a fiscal year) and includes a set of budgeted financial statements.
- B) A budget provides a frame of reference, a set of specific expectations against which actual results can be compared.
- C) Variations from the budget will be investigated. If necessary, corrective action follows investigation.
- D) All of the above.

2. Sales in December 2021 were \$960 000. Projected sales for the first quarter of 2022 are:

January \$1,080,000

February \$1,300,000

March \$1,280,000

Sales are 20% cash and 80% on credit. Debtors pay in the month following the sale. Refer to the table above. Calculate the total cash collected from sales for the month of March.

- A) \$1,296,000
- B) \$1,456,000
- C) \$1,216,000
- D) \$256,000



Multiple Choice Questions

3. The Howell Company has prepared a sales budget of 43,000 finished units for a 3-month period. The company has an inventory of 11,000 units of finished goods on hand at December 31 and has a target finished goods inventory of 19,000 units at the end of the succeeding quarter.

It takes 4 gallons of direct materials to make one unit of finished product. The company has an inventory of 66,000 gallons of direct materials on December 31 and has a target ending inventory of 56,000 gallons at the end of the succeeding quarter. How many gallons of direct materials should Howell Company purchase during the 3 months ending March 31?

- A) 510,000 gallons
- B) 260,000 gallons
- C) 194,000 gallons
- D) None of the above is correct



Demonstration Question

Question 1

Records of Recycling Ltd contain the following data:

Opening bank balance \$35,000

	Sales	Purchases
May	\$60,000	\$20,000
June	70,000	30,000
July	80,000	25,000

All sales are on credit. Records show that 80% of the customers pay during the month of the sale, 15% pay in the month after the sale and the remaining 5% pay in the second month after the sale. Purchases are all paid the following month at a 1% discount. Cash disbursements for operating expenses in July were \$8,000.

REQUIRED:

Prepare a cash budget for July.

Solution

Cash Budget for July

Cash Receipts

July sales (80% of \$80 000)	\$64,000
June sales ((15% of \$70 000)	\$10,500
May sales (5% of \$60 000)	<u>3,000</u>
Total cash receipts	<u>\$77,500</u>

Cash Payments

June purchases (\$30 000 × 99%)	\$29,700
Operating expenses	<u>8,000</u>
Total cash payments	<u>\$37,700</u>

Increase in cash	39,800
Opening cash balance	<u>35,000</u>
Closing cash balance	<u>\$74,800</u>

Practice Question

Question 1

Jackson BottleYard, a recycling glass company, has supplied the following information in relation to their actual sales in 2017 and planned sales for the first quarter of 2018.

		Cash \$	Credit \$	Total \$
2017 Actual Sales for	November	25,000	25,000	50,000
	December	35,000	40,000	75,000
2018 Estimated Sales for	January	15,000	20,000	35,000
	February	18,000	30,000	48,000
	March	22,000	40,000	62,000

Past records indicate credit sales are collected as follows:

- 60 percent in the month of sale
- 40 percent in the month following the sale

REQUIRED:

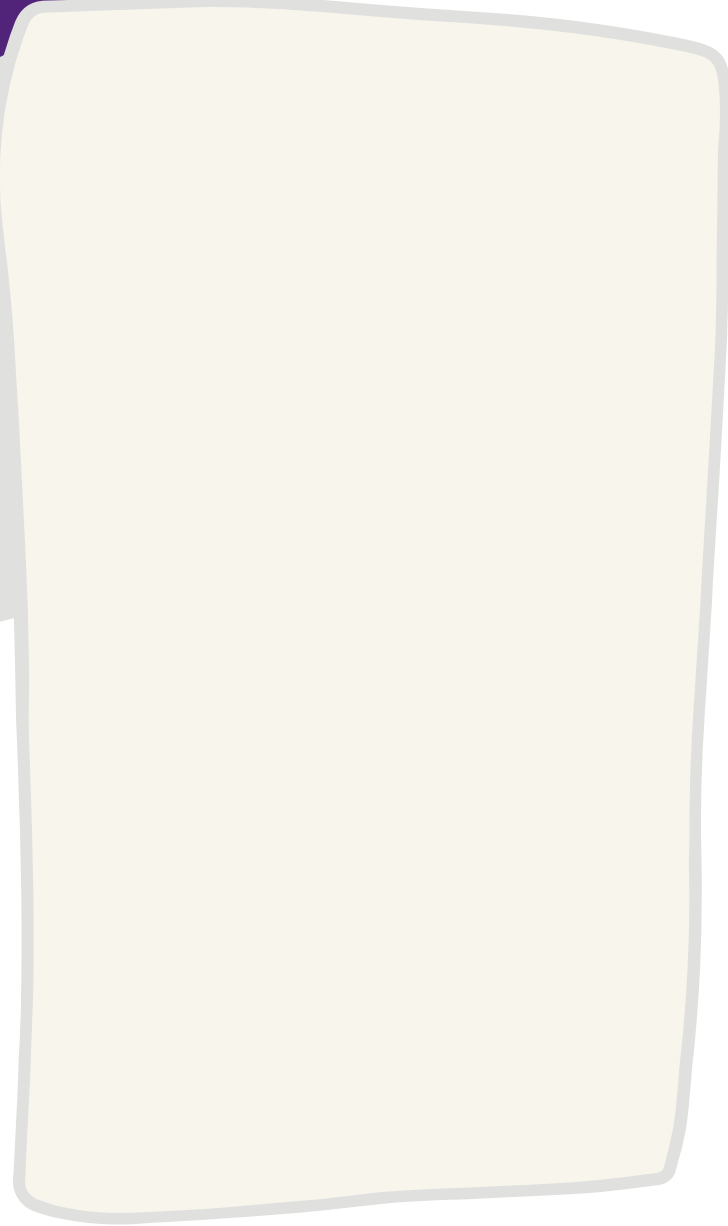
What are the cash receipts from accounts receivable for the months of January to March?

Solution

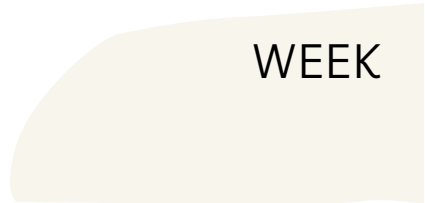
	January	February	March
Credit sales			
Credit sale cash receipts			
from current month sales (60%)			
from previous month sales (40%)			
Total cash receipts from accounts receivable			

WEEKLY NOTES

TO DO

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WEEK

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NOTES

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NEXT WEEK

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Notes

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